# Strict Horizons and a Free Quotient

## Smaller explicit Diophantine equations for ZFC consistency

AI-assisted mathematical research

Prepared for Vladimir Reshetnikov and the ProveIt project

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**Abstract**

We give a fully specified quartic polynomial in 236 unrestricted integer variables, with 4,876 collected monomials and coefficient height 7,968, whose absence of integer zeros is provably equivalent in ZFC to Con(ZFC). It is the sum of the squares of 173 explicitly printed quadratic polynomials, each of coefficient height at most 64. A denser companion has 235 variables. The direct construction uses the repository's numerical proof calculus and contains no large machine coefficient or specialization of a universal Diophantine polynomial.

The principal new step chooses the proof horizon *strictly* above all local schema parameters and derived labels. Five range certificates then share an existing binomial denominator, removing an entire exact power request and its auxiliary data. A self-power majorant $F = t^t$, a linear CRT radius $F$, and an affine signed radix $q = 4z + 1$ give further compatible savings. Both directions are proved, including sign recovery before binomial decoding and simultaneous cofinal witness choices.

A separate compact construction moves an arithmetic scale from a quotient coordinate into its Pell base. It gives ten-positive-variable polynomials with 83 binary operations and degree 312, or 89 operations and degree 278. These retain the complete large fixed coefficient tuple and its inherited machine calibration. Exact algebraic audits and reconstructors accompany all four equations. The new unbounded arguments are ordinary proofs; the inherited logical and Pell results are identified, and no new complete proof-assistant formalization or global minimality claim is made.

Repository comparison snapshot:
`7a8da414376a80b90ecf799f37ceb26fb9ac674e`.
The equations are fixed; no consistency assumption is used to prove their calibration.

# Contents

**1 The result and the choice of objective** **3**
1.1 Exact improvements over matched incoming equations . . . . . 3
1.2 What is new, and what is inherited . . . . . 4

**2 The calibrated direct interface** **4**
2.1 Pell requests and signed bases . . . . . 4

**3 Strict horizons remove a whole power certificate** **5**

**4 A self-power and a linear CRT radius** **6**
4.1 Label multiplicities at the self-power bound . . . . . 6
4.2 The radius need only be linear in the carrier . . . . . 7
4.3 The two binomial prime-power facts . . . . . 7

**5 An affine radix, with signs recovered in order** **8**
5.1 Bootstrap before decoding the carrier . . . . . 8
5.2 Recover the exact binomial at either sign of the radix . . . . . 9
5.3 The remaining digits, strict derived range, and proof flow . . . . . 9

**6 Simultaneous constructive completeness** **10**

**7 From the source to the literal equations** **11**
7.1 Finite verification and its limits . . . . . 11

**8 A free quotient in the compact Pell encoding** **12**
8.1 The change of arithmetic architecture . . . . . 12
8.2 Complete polynomial specification . . . . . 12
8.3 Ordered soundness argument . . . . . 13
8.4 Constructive converse and exact degree . . . . . 15

**9 A cofinal Pell experiment that did not improve the endpoint** **16**

**10 Use and incorporation** **16**

**A Complete principal quadratic residuals** **17**

**B Artifact identities and comparison scope** **23**

# 1 The result and the choice of objective

The recommended small-coefficient equation is

$$P_{236}(x_0,\dots,x_{235}) = \sum_{j=0}^{172} R_j(x_0,\dots,x_{235})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{236}. \tag{1}$$

Every $R_j$ is printed in Appendix A. This is a literal definition by small integer coefficients, not a prescription to compile an unspecified machine. The archive contains the same residuals in `direct/principal236/equation.json`, their complete collected quartic, readable exports, and exact transformation certificates.

**Theorem 1.1** (Fixed consistency equations)**.** *For the explicitly supplied polynomials $P_{236}$ and $P_{235}$,*

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{236}\ P_{236}(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)},$$

*and the corresponding statement holds with* 235 *variables and $P_{235}$. The two compact polynomials of Section 8 satisfy the analogous equivalence over ten strictly positive integers.*

Here Con(ZFC) means the arithmetic statement that no finite syntactic ZFC refutation exists, with the repository's specified proof system. The proofs establish an equivalence of *existence of certificates*, without assuming consistency. The theorem does not assert that ZFC proves its own consistency or the actual unsolvability of these equations. It is an ordinary mathematical theorem obtained by composing the new arithmetic proofs with the identified inherited calibration. The verification programs check finite algebraic data, not this entire unbounded theorem.

## 1.1 Exact improvements over matched incoming equations

| Unrestricted-integer quartic | Variables | Rows | Terms | Height | Row height |
|---|---|---|---|---|---|
| Factorial-free parent, principal | 242 | 178 | 5,018 | 7,968 | 64 |
| **New principal** | **236** | **173** | **4,876** | **7,968** | **64** |
| Factorial-free parent, dense | 241 | 177 | 12,774 | 7,968 | 64 |
| **New dense companion** | **235** | **172** | **12,566** | **7,968** | **64** |

All four rows have exact total degree four. Thus each new equation saves six variables and five residuals against its matched parent, while also reducing support without increasing either coefficient height. The principal has 142 fewer collected monomials; the dense companion has 208 fewer. The denser equation is a different tradeoff from the principal: one coordinate is removed at a substantial increase in support.

| Ten-positive-variable circuit | Operations | Exact degree | Comparison |
|---|---|---|---|
| Incoming plain point | 84 | 328 | retained predecessor |
| **New plain point** | **83** | **312** | both improve |
| Incoming corrected point | 89 | 288 | retained predecessor |
| Incoming corrected point | 92 | 278 | retained predecessor |
| **New corrected point** | **89** | **278** | dominates these two |

An operation is one binary addition, subtraction, or multiplication in the live straight-line circuit. Reusing a computed value does not cost another operation. The compact equations use a large fixed coefficient tuple, whereas the direct quartics use only the displayed small coefficients. These metrics must not be conflated. The repository already contains equations with fewer variables and other favorable tradeoffs [1]. We have neither a definition-independent smallest equation nor a lower bound proving any of these endpoints optimal.

### 1.2 What is new, and what is inherited

The direct arithmetic contribution is the combination of a strict proof horizon, a shared range denominator, self-power growth, a smaller CRT radius, and a signed affine radix with a noncircular sign proof. The compact contribution is a fresh placement of the quotient scale in the Pell construction. These are concrete improvements over the inspected sources. We do not claim to have established worldwide priority.

The factorial-free binomial CRT compiler and its two complementary prime-power lemmas are inherited from the incoming factorial-free submission [2]. Centered common remainders, the semantic Pell anchor, signed factorization, and the integral balancing operations are also retained and credited [3, 4]. The signed-digit idea is adapted from the Shared Factorial Error submission [5]. The compact comparison and its native calibration come from the incoming Lucas-correction and cofinal-witness submissions [6, 3].

## 2 The calibrated direct interface

The direct route starts from the fixed numerical calculus $C_{222}$. Its finitely described local rules have one conclusion and at most two premises. The truth label is 0 and the refutation label is 1. The retained local polynomial tests whether a row is a valid instance and enforces two adjacent beta-code residues. A polynomially derived local label sum $J$ bounds every label used by that row.

The logical endpoint is the composition

$$C_{222} \text{ derives } 1 \quad \longleftrightarrow \quad \text{a finite ZF refutation exists} \quad \longleftrightarrow \quad \text{a finite ZFC refutation exists.}$$

The named retained results are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their pinned source files, syntax calibration, and ordinary bridge proofs are included in the dependency tree [7]. Although the same calculus is used elsewhere in proofs about machines, the present direct polynomial compiles its numerical rules directly. It does not encode a run of K215 and does not use Jones's universal expression.

Let $t$ denote the number of padded local rows. The actual CRT representatives are $C$ and eight values $A_a, A_b, A_c, A_d, A_S, A_{S'}, A_u, A_v$. In the raw source several stored coordinates are these values minus one. The literal auditors use those shifts explicitly; semantic names in this report refer to actual values. The derived label $J(\boldsymbol{A})$ is evaluated from those representatives. It is *not* an extra independent CRT column.

For a bounded decoded local tuple $\boldsymbol{w}$, the retained nonnegative polynomial $\mathcal{L}$ has the first of the fixed majorants below; the second is a separate bound on the globally encoded derived label:

$$0 \le \mathcal{L}(i, \boldsymbol{w}), J(\boldsymbol{w}) < (2 + t + x + y + F)^E, \qquad E = 8{,}388{,}608, \tag{2}$$

and $J(\boldsymbol{A}) \le H^\ell$ with $\ell = 4{,}194{,}304$. where $H = 2 + C + \sum A_j$ for nonnegative representatives. Its precise arithmetic graph and the proofs of these generous bounds are retained. Changing a range from $\le t$ to $< t$ preserves these upper bounds.

### 2.1 Pell requests and signed bases

For an integer center $A$, define Pell polynomials by

$$X_0 = 1, \quad Y_0 = 0, \qquad X_{k+1} = AX_k + (A^2 - 1)Y_k, \quad Y_{k+1} = X_k + AY_k,$$

with $X_{-k} = X_k$ and $Y_{-k} = -Y_k$. The retained square frame, pooled index residues, and one semantic anchor force these exact values at the requested signed indices. They give a frame $W \ge 258$ that strictly

bounds all actual base, index, and output atoms and satisfies $|A| > W^W$. The anchor uses the *encoded* derived label $J(\boldsymbol{A})$; the small decoded local label is a different value. No step here substitutes the latter for the former.

For a signed base $n$ the power congruence uses

$$T_n = 2An - n^2 - 1.$$

The unit property of $n$ modulo $T_n$, the Pell recurrence, and the absolute error bound imply the following inherited consequence:

$$|n| \geq 2 \quad \Longrightarrow \quad \text{a certified request } (n, k, m) \text{ forces } k \geq 0, m = n^k. \tag{3}$$

For negative $k$ the congruence would force $mn^{-k} = 1$, impossible for $|n| \geq 2$. For nonnegative $k$ it forces $m = n^k$, since the possible error is smaller in absolute value than $|T_n|$. This statement allows negative bases. That feature is essential below. Variable powers in the semantic discussion are conclusions of these ordinary quadratic certificates, not operations in the final polynomial.

## 3 Strict horizons remove a whole power certificate

In the matched parent, four local schema parameters and the derived label have bounds $a, b, c, d, J \leq t$. Their binomial ranges use lower index $t + 1$ and denominator $q^{t+1}$. The index column already has bound $C_{\text{local}} < t$ and denominator $q^t$.

Choose instead

$$t = 16 + h^2, \qquad a, b, c, d, J < t. \tag{4}$$

Completeness is preserved: a finite proof, its finitely many row parameters, and its derived labels can all be fixed *before* choosing a larger horizon. The fixed truth-padding row also has bounded parameters and derived label. Taking a sufficiently large allowed horizon dominates this finite set; it does not create a dependence on the later CRT representatives or Pell witnesses.

The five range requests now use $\binom{A}{t}$, with the appropriate actual upper representative $A$. They share the existing $q^t$ denominator and its normalization ratio. Hence we remove the request for $q^{t+1}$, its one-step Pell addition graph, eight private source coordinates, and the duplicate denominator-ratio equation. Some of those coordinates would already disappear in monic elimination; the final count is obtained from the emitted graph, not from adding raw removals.

The surviving 17 requests are exactly as follows. Identical indices share their principal Pell pair.

| Base | Index or indices | Purpose |
|---|---|---|
| $t$ | $t$ | $F = t^t$ |
| $B$ | $x$ | $N = B^x$ |
| $BH$ | $x$ | $q = (BH)^x$ |
| $q + 1$ | $F + t$ | exact binomial carrier |
| $q$ | $t,\ y + 1,\ x$ | three denominator groups |
| $q + 1$ | $A_a, A_b, A_c, A_d, A_S, A_{S'}, A_u, A_v, C$ | nine independent upper representatives |
| $q + 1$ | $J(\boldsymbol{A})$ | derived-label range and anchor |

There remain fourteen principal indexed Pell pairs and five base moduli. The frame deletes the obsolete output and the obsolete $t + 1$ index. Since the first base is now the existing index $t$, one duplicate square atom is also removed. Every required atom remains bounded. The new constructive frame is rebuilt; no equality with an old frame witness is claimed.

# 4 A self-power and a linear CRT radius

The selected arithmetic choices are

$$F = t^t, \qquad R_{\mathrm{CRT}} = F, \qquad B = \binom{F+t}{t}, \qquad y = F(2v+1)^2. \tag{5}$$

Only the first equality needs a power certificate, at the already present index $t$. The other displayed expressions describe the mathematical meanings of literal quadratic registers and binomial digits.

## 4.1 Label multiplicities at the self-power bound

For each nonnegative label put $W_j = 1 + jF$. The commutative proof certificate eventually gives

$$\prod_{i<t} W_{v_i} = (1+F) \prod_{i<t} W_{p_i} W_{q_i}. \tag{6}$$

Unused premise slots have label zero and contribute one.

**Lemma 4.1** (Self-power separation)**.** *Let $t \geq 2$, $F = t^t$, and suppose all labels in (6) are at most $t$. Then for every positive label $j$, its number of conclusion occurrences minus its number of premise occurrences is one for $j = 1$ and zero otherwise.*

*Proof.* Discard all factors of label zero. If the remaining factor counts differ, their minimum $r$ is at most $t$, because the conclusion side has at most $t$ factors. The larger product is at least $(F+1)^{r+1}$ and the smaller at most $t^r(F+1)^r$. But $F + 1 > t^t \geq t^r$, a contradiction. Thus both counts are the same $r \leq t$. For a product $U(z) = \prod_{i=1}^{r}(1 + a_i z)$ with $1 \leq a_i \leq t$, every coefficient $u_j$ is positive and

$$1 \leq u_j \leq \binom{r}{j} t^j \leq r^{r-j} t^j \leq t^r \leq F.$$

Therefore corresponding coefficients of the two product polynomials differ by at most $F - 1$. This difference bound remains strict even when an individual coefficient equals $F$. If a nonzero difference polynomial vanished at $F$, take its lowest nonzero coefficient, divide the value by the corresponding power of $F$, and reduce modulo $F$. This would force a nonzero integer of absolute value less than $F$ to be divisible by $F$, a contradiction. Consequently the products agree in $\mathbb{Z}[z]$. Their distinct factors $1 + jz$ have distinct roots $-1/j$, so unique factorization recovers exactly the claimed multiplicities. □

The lemma is deliberately stated for the slightly larger range $j \leq t$; the new strict ranges are more than sufficient. It avoids the incorrect claim that every individual coefficient is strictly smaller than $t^t$.

**Lemma 4.2** (Single-conclusion flow)**.** *If the factors in (6) come from valid single-conclusion Horn rules and truth is derivable, then label 1 is derivable.*

*Proof.* Take the closure of truth under the finitely many listed rule instances. If label 1 is outside, any rule with conclusion outside has at least one premise outside. Summing conclusions minus premises over outside labels is at most zero. Lemma 4.1 makes that sum one, a contradiction. This argument permits repeated premises and cycles; a cycle cannot supply a positive net conclusion. □

### 4.2 The radius need only be linear in the carrier

**Lemma 4.3.** *For $t \geq 1$, $\Lambda_t = \operatorname{lcm}(1, \ldots, t)$ satisfies $\Lambda_t \leq 4^{t-1}$. For $t \geq 16$,*

$$\Lambda_t^2 \leq 16^{t-1} < 16^t \leq t^t = F.$$

*Proof.* Let $k = \lceil n/2 \rceil$. A prime power present in $\Lambda_n$ and missing from $\Lambda_k$ is greater than $k$ and at most $n$; there is at most one additional power of that prime. Its contribution to $\binom{n}{\lfloor n/2 \rfloor}$ has valuation at least one, by the floor-function formula. Thus

$$\Lambda_n \mid \Lambda_k \binom{n}{\lfloor n/2 \rfloor}.$$

For $n \geq 2$ the central binomial coefficient is at most $2^{n-1}$. For odd $n$ this follows from its equal central neighbor; for even $n = 2m$ use $\binom{2m}{m} = 2\binom{2m-1}{m-1}$ and the odd case. Induction gives $\Lambda_n \leq 4^{k-1}2^{n-1} \leq 4^{n-1}$, with initial value $\Lambda_1 = 1$. Squaring and using $t \geq 16$ proves the final chain. □

The matched parent used the radius $F^2$. Its generalized binomial CRT theorem needs only $R_{\mathrm{CRT}} \geq \Lambda_t^2$. The lemma therefore licenses $R_{\mathrm{CRT}} = F$, deleting a multiplication without adding a certificate for a factorial or for an lcm.

### 4.3 The two binomial prime-power facts

For clarity we include the inherited facts that make the radius argument sufficient. They have different roles in the two directions.

**Lemma 4.4** (Every prime has a compatible row)**.** *For $R, t \geq 1$, put $B = \binom{R+t}{t}$. If $p^e \| B$, then $p^e \mid R + i$ for some $1 \leq i \leq t$.*

*Proof.* Let $v = \max_i v_p(R + i)$. The factorial valuation for $B$ is a sum of zero-one terms

$$e = \sum_{a \geq 1} \left( \left\lfloor \frac{R+t}{p^a} \right\rfloor - \left\lfloor \frac{R}{p^a} \right\rfloor - \left\lfloor \frac{t}{p^a} \right\rfloor \right).$$

For $a > v$ the interval contains no multiple of $p^a$. Such a power also exceeds $t$, so that summand is zero. Therefore $e \leq v$, and a row attaining the maximum is compatible. □

**Lemma 4.5** (Every row has a large selector)**.** *If $R \geq \Lambda_t^2$, then for each $1 \leq i \leq t$ there is $p^e \| \binom{R+t}{t}$ with $p^e \mid R + i$ and $p^e > t$.*

*Proof.* First observe the classical divisibility

$$t \binom{t-1}{i-1} \mid \Lambda_t.$$

Indeed its valuation at level $p^a$ is $\lfloor t/p^a \rfloor - \lfloor (i-1)/p^a \rfloor - \lfloor (t-i)/p^a \rfloor$, which is zero or one and vanishes for $p^a > t$. This gives valuation at most $\lfloor \log_p t \rfloor$.

Since $R + i > \Lambda_t^2$, some prime has $v = v_p(R + i) > 2a$, where $a = v_p(\Lambda_t)$. Then $p^v > t$; for $j \neq i$, $v_p(R + j) = v_p(j - i)$. Writing $d = v_p(t\binom{t-1}{i-1}) \leq a$, the full valuation in $B$ is $e = v - d$. Thus $a < e \leq v$, so $p^e > t$ and $p^e \mid R + i$. □

**Lemma 4.6** (Binomial range)**.** *If $A \in \mathbb{N}$, $k, e \geq 1$, and $p^e \mid \binom{A}{k}$, then $A \bmod p^e < k$, with $\binom{A}{k} = 0$ when $A < k$.*

*Proof.* The case $A < k$ is immediate. Otherwise, if no factor of $A - k + 1, \ldots, A$ is divisible by $p^e$, all valuation terms from level $e$ onward are zero. The remaining zero-one terms total at most $e - 1$, a contradiction. Hence $A - r$ is divisible by $p^e$ for some $0 \leq r < k$, and the ordinary residue is less than $k$. □

For soundness, the constraints

$$B \mid C + R + 1, \qquad B^x \mid \binom{C}{t}, \qquad B^x \mid \binom{A_j}{b_j + 1}, \qquad B^x \mid \mathcal{L}(C, \boldsymbol{A})$$

select each row using Lemma 4.5. Modulo its $p^{xe}$, Lemma 4.6 decodes bounded values. The raw index equation and $p^e > t$ identify the selected row exactly. If $0 \leq \mathcal{L} < 2^x \leq p^{xe}$, its divisibility forces zero.

For completeness, use Lemma 4.4 at *every* prime of $B$. Prescribe a valid compatible row and its tuple modulo $p^{2xe}$. CRT combines these conditions modulo $B^{2x}$. In a range with lower index $k \leq x$, its falling factorial has a factor divisible by $p^{2xe}$. Division by $k!$ loses at most $v_p(k!) < x \leq xe$, so $p^{xe}$ still divides the binomial. This supplies all ranges and the local congruence. A selector for each row alone would not prove this converse; the all-primes lemma is necessary.

# 5 An affine radix, with signs recovered in order

Replace the parent's odd-square radix by

$$q = 4z + 1, \qquad z \in \mathbb{Z}. \tag{7}$$

This permits a linear elimination after all other transformations. The residue class excludes $q = 0, -1, -2$, so $|q + 1| \geq 2$ before positivity is known. The weaker affine odd class $2z + 1$ would admit $q = -1$, whose numerator base is zero. The Shared Factorial Error construction excludes unit radices using a separate factorial equation; that equation is absent here. The modulus-four choice is what permits the transfer of its signed-digit idea to this graph.

## 5.1 Bootstrap before decoding the carrier

The known positive base $t \geq 16$ first gives $F = t^t > 2^{32}$. The odd-square beta multiplier gives $y \geq F > 0$. All numerator powers have base $q + 1$ with absolute value at least two, so their actual upper indices are nonnegative. In particular $C, A_j \geq 0$ and $H \geq 2$. The raw index equation is

$$Bh_B = C + F + 1.$$

It excludes $B = 0$, hence $|BH| \geq 2$. The certified request $q = (BH)^x$ now forces $x \geq 0$, even if $B$ is provisionally negative. The endpoints are

$$x = F + 1 + (y + 1)u_0, \qquad x = 1 + [1 + (t + 1)y]u_t. \tag{8}$$

The last equation excludes $x = 0$. If $u_0 \leq -1$ the first gives $x \leq F - y \leq 0$, impossible. The last endpoint then has $u_t \geq 1$. Consequently

$$x \geq F + 1, \qquad x \geq 2 + (t + 1)y \geq 2 + (t + 1)F. \tag{9}$$

With $Q = |q|$ we obtain

$$Q = |BH|^x \geq 2^x, \qquad |B| \leq |BH| < Q/2.$$

In particular $|q| \geq 2$, so its denominator requests are genuine signed powers. No claim $B > 0$ or $q > 0$ was used.

### 5.2 Recover the exact binomial at either sign of the radix

Put $n = F + t$ and $c_j = \binom{n}{j}$. For $0 \leq j \leq t$,

$$c_j \leq (F + t)^t \leq (F^2 + F)^t \leq 2^{t(F-1)}.$$

The last elementary inequality holds for $F \geq 7$; it is much weaker than the actual size of $F$. Together with (9) it gives

$$\frac{\max_{j \leq t} c_j}{Q - 1} < \frac{1}{32}, \qquad c_t < Q/32.$$

The signed truncated polynomial

$$r_* = \sum_{j=0}^{t-1} c_j q^j$$

is congruent to $(q + 1)^n$ modulo $q^t$ and satisfies $|r_*| < Q^t/32$, regardless of the sign of $q$.

The common denominator is $M = q^x$. Every range or exact digit has a denominator $D$ and integral ratio $v_D$ with $Dv_D = M$. A scaled row has the form

$$v_D U = MK + \rho.$$

As $M \neq 0$, also $v_D \neq 0$; the row shows $v_D \mid \rho$. Thus $r = \rho/v_D$ is an integer and $U = DK + r$. The common norm is

$$M^2 = 1 + 4 \sum_{j=1}^{11} \rho_j^2 + \alpha^2 + \beta^2 + \gamma(\gamma + 1). \tag{10}$$

Every last term is nonnegative for integer witnesses. It follows that $|r| < |D|/2$. In particular the carrier digit's remainder and $r_*$ are congruent modulo $q^t$ and differ in absolute value by less than $(1/2 + 1/32)|q^t|$. They are equal.

Subtracting the low polynomial from $(q + 1)^n$ and dividing by $q^t$ uses the literal carrier-digit quotient $K = B + qh_{\text{high}}$:

$$(q + 1)^{F+t} = q^t(B + qh_{\text{high}}) + r.$$

It therefore gives $B \equiv c_t \pmod{q}$. But $|B - c_t| < Q/2 + Q/32 < Q$, so

$$B = \binom{F + t}{t} > 0, \qquad q = (BH)^x > 0.$$

The signs are consequences of the exact digit, rather than premises of its decoding. The remaining arithmetic may now use $B \geq 2$ and $N = B^x > 0$.

### 5.3 The remaining digits, strict derived range, and proof flow

For ordinary representatives $a \geq 0$ and lower indices $j \leq k \leq x$, $\binom{a}{j} \leq \max(1,a)^k \leq H^x$. For the derived label use (2) and $\ell t < x$ to obtain the same bound. Since $q = B^x H^x$, all remaining genuine low remainders are less than their denominator divided by 32. The norm selects them uniquely. There are now three denominator groups: $q^t$, $q^{y+1}$, and $q^x$. The first serves the carrier, the index range, and all five strict ranges.

The derived range gives, at a selected prime power, $J(\boldsymbol{A}) \equiv r \pmod{p^{xe}}$ with $0 \leq r < t$. Column congruences give $J(\boldsymbol{w}) \equiv r$ modulo the same power. The inherited local bound gives $0 \leq J(\boldsymbol{w}) < 2^x \leq p^{xe}$, hence $J(\boldsymbol{w}) = r < t$. This is the required justification for a derived column; treating it as an independent representative would not suffice.

The bounds $y \geq F$, (9), and $t \geq 16$ give $2 + t + x + y + F < 2x$ and $x > 2^{32}$. The elementary dyadic estimate $2^{23} \log_2(2x) < x$ therefore yields

$$0 \leq \mathcal{L} < (2x)^{2^{23}} < 2^x.$$

Lemma 4.3 licenses radius $F$; the binomial CRT argument recovers every bounded valid local row.

The algebraic beta choice permits $y = F$. In that case the first endpoint has residue zero modulo $y + 1$, so its initial accumulator is zero. Each local update

$$S_{i+1} W_{v_i} = S_i W_{p_i} W_{q_i}$$

propagates zero because all weights are positive. The last endpoint has residue one, a contradiction. Thus an actual zero has $y > F$, initial accumulator $F + 1$, and final accumulator one. Multiplying the updates gives (6); label separation and single-conclusion flow give a finite $C_{222}$ refutation. The retained logical calibration gives a finite ZFC refutation.

## 6 Simultaneous constructive completeness

Start with a finite calibrated $C_{222}$ refutation, expand it to a proof tree, and fix its finite rule parameters and label data. The accumulator and beta-quotient values will be chosen afterward. Choose a sufficiently large *even* $h$ so that $t = 16 + h^2$ strictly exceeds the tree size, all schema parameters, all local derived labels, and those of the fixed truth-padding row. Set $F = t^t$ and form the pending-multiset accumulators using weights $1 + jF$. Truth padding gives exactly $t$ rows. This chooses the horizon before all arithmetic witnesses.

The parity choice is essential for this self-power:

$$v_2(F) = t\, v_2(t) \geq t > v_2(t!).$$

Let $O$ be the odd part of $t!$ and choose an arbitrarily large odd multiple $w$ of $O$. Set $y = Fw^2$, large enough to exceed $F$ and all pending accumulators. Then $y$ is an even multiple of $t!$. The beta moduli $1 + (i+1)y$, $0 \leq i \leq t$, are pairwise coprime: a common prime for two of them would divide their index difference, then $y$, then one of these numbers, an impossibility. They are odd. CRT supplies a common beta code $x$; adding their odd product lets us choose $x$ even and large. The endpoint quotients and all local beta quotients meet their required ranges. The odd $w$ is exactly $2v + 1$ for an integer $v$.

Define $B = \binom{F+t}{t}$ and $N = B^x$. The binomial CRT construction supplies the nine representative classes modulo $N^2$. For each $1 \leq i \leq t$, $v_2(F) > \lfloor \log_2 t \rfloor$ gives $v_2(F + i) = v_2(i)$, so $B$ is odd. Thus each representative class has freely selectable parity. Choose the dominant schema representative $a$ odd in $[2N^2, 4N^2)$ and the other eight representatives positive even and at most $2N^2$.

The retained shell-label construction has $J(\boldsymbol{A}) \equiv 1 + a + b \pmod 2$ and $J(\boldsymbol{A}) \geq 2a^2$. Hence the semantic anchor is even and dominates every other index. Indeed $F + t \leq B < N^2$, $x, y + 1, t < N^2$, while the anchor is at least $8N^4$. The derived strict range is preserved because evaluating its polynomial respects the CRT congruences modulo $p^{2xe}$; division by $t!$ loses fewer than $x$ powers of $p$.

Now $H = 2 + C + \sum A_j$ is odd. With $x$ even, $q = (BH)^x$ is an odd square, hence $q \equiv 1 \pmod 4$ and $z = (q-1)/4$ is an integer. These canonical witnesses satisfy the new affine radix. The true normalized remainders obey $0 \leq \rho_j < M/32$, and

$$4 \sum_{j=1}^{11} \rho_j^2 < \frac{44}{1024} M^2 < M^2.$$

The nonnegative integral gap $g = M^2 - 1 - 4 \sum \rho_j^2$ has the form $\alpha^2 + \beta^2 + \gamma(\gamma+1)$. To see this, represent $4g + 1$ by three squares; it is 1 modulo 4, so exactly one square is odd. Writing the two even roots as $2\alpha, 2\beta$ and the odd one as $2\gamma + 1$ gives the claimed identity.

The inherited constructive frame and semantic-anchor theorem supply fresh simultaneous Pell witnesses for the seventeen genuine power requests. Removing a duplicate frame atom does not obstruct that construction. All other certified rows now have their intended values. Thus every finite ZFC refutation gives a zero of the changed arithmetic source, and the exact integral transformations give a zero of each final quartic.

Factorials occur here only to prove existence of a suitable multiple in the converse. No unknown in the new certificate is required to equal $t!$. In particular the change from $[2(t+1)]^t$ to $t^t$ must use the even cofinal choice above; copying an odd-$h$ choice from an intermediate $(t+1)^t$ variant would be wrong.

# 7 From the source to the literal equations

The final source retains only ordinary quadratic residuals. The signed Horn factorization uses the exact identity

$$\left(\prod L_i^2\right)(U^2 + V^2) = \left(\prod L_i\right)^2 (U^2 + V^2)$$

and an analogous identity for the guarded local rule. The literal factorization certificate is independently checked. Then 198 monic pivots are eliminated. A pivot $x - f(\boldsymbol{y}) = 0$ permits integral substitution in every remaining row, and its inverse restores $x = f(\boldsymbol{y})$; each chosen substitution preserves degree at most two. Unused coordinates may be dropped. Recorded integral translations give the 239-variable, 176-row source.

Both endpoints apply six integral coordinate shears and five preliminary row operations. Their coefficients and order are in the certificate. Each shear is invertible over the integers, and a row replacement $R_i \leftarrow R_i + kR_j$ with $i \neq j$ has the inverse $R_i \leftarrow R_i - kR_j$. Three affine projections give the principal; four give the dense companion. The final balancing uses respectively 33 and 35 further row operations. One parent operation that used the deleted successor-power row is omitted explicitly. The mappings are transported by original raw-row identities, not by guessed row offsets.

These operations preserve the existence of an integer zero of the simultaneous residual system. They need not preserve the numerical value of a sum of squares away from its zero set. Over the integers, $\sum R_j^2 = 0$ is equivalent to every $R_j = 0$. The collected expansions have nonzero quartic terms and all coordinates occur, so their degrees and variable counts are exact.

## 7.1 Finite verification and its limits

The release separates the following checks from the ordinary proofs:

1. Exact reconstruction of the parent-to-source edit, with a complete census of changed and deleted rows.
2. An independent raw-interface audit of all seventeen requests, fourteen principal norms, five base moduli, 34 deduplicated frame atoms, eleven cap rows, six anchor-assembly rows, all ten range identities, and the common normalized remainder budget.
3. Independent Horn factorization, all 198 monic projections, translations, endpoint maps, every collected quartic coefficient, and coordinate liveness.
4. Fresh byte-identical reconstruction of the source and both direct endpoints, and of the compact circuits and fixed coefficients.
5. Exact free-polynomial cut identities for every compact gate, complete leading homogeneous forms, and deliberate corruption controls. The leading forms prove the exact degrees; random evaluations would not be sufficient.

Normal Python and optimized Python runs are both supplied; the audits do not rely on removable `assert` statements. Negative controls alter actual certificates or gates and are required to be rejected. Checksums fix the artifacts, but are not evidence for the unbounded arithmetic lemmas.

The source calibration and unchanged Pell lemmas are inherited mathematical dependencies. Their pinned files, ordinary proof sources, and original notices are retained. The archive is not a standalone Lean or Rocq project closing every imported theorem, and no fresh full repository proof-assistant build was performed. The principal new formalization work still needed is the strict horizon, self-power, signed-radix, and free-quotient proof layer.

# 8 A free quotient in the compact Pell encoding

This section gives two additional equations in ten strictly positive integer variables. Their exact arithmetic costs and total degrees are 83/312 and 89/278. The second improves the earlier 89/288 and 92/278 points in cost and degree. The first improves the earlier 84/328 point in both metrics. Other tradeoffs remain relevant. These are ordinary proofs composed with the retained native/refutation calibration, not new proof-assistant certificates or global optimality claims.

## 8.1 The change of arithmetic architecture

The incoming constructions used $Y = qs$ and $c = kY + \eta$. The factor $q$ then propagated through $U = ic^2$, making the strong Pell part expensive in degree. We instead take

$$Y = s, \qquad \lambda = q,$$

and put $\lambda$ into the two Pell bases. This retains the needed divisibility while lowering the degree of $c$. The complete denominator is recovered only after exponential projection. The positive shift $T + \alpha$ is restored: it supplies a direct early proof of $b > c$ when $Y$ is no longer divisible by $q$.

## 8.2 Complete polynomial specification

The independent variables are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed tuple $(\beta, K_0, M_C, M_F)$ is the complete directly indexed BFS943 radix67 tuple retained with this component. Its largest port has 592,884 bits. Every byte and its finite reconstruction are included;

a port name is not an extra unknown. In particular $\beta \geq 31$ is odd and $K_0 > 0$, $0 < M_C < \beta < M_F < 2\beta$. Set

$$h = \beta J, \quad q = h + 1, \quad C = q - F - \alpha, \quad Y = s.$$

Choose either the plain chart or the corrected chart:

| | $W_0$ | $X$ | $D$ |
|---|---|---|---|
| plain | $qw$ | $q^2 w$ | $q^7$ |
| corrected | $w$ | $qw$ | $6q^7$ |

Both use the following polynomial abbreviations:

$$\begin{aligned}
k &= \eta + \zeta, & c &= kY + \eta, & & \\
A &= qY(X+1) + 1, & \Delta &= A^2 - 1, & H &= 4A - 5, \\
v &= Xq^2Y^2, & E &= 1 + v(v+1)k^2, & & \\
R &= [h(q-F) + \alpha](q^2-1) + (M_C + qM_F)J, & & & & \\
B &= (K_0 + W_0)C - F, & U &= ic^2, & & \\
a &= \Delta U^2 E, & F_0 &= 1 + a, & G &= EF_0, \\
F_s &= 1 + 4aF_0, & Q &= \Delta(F_s - 1), & & \\
b &= c(T + \alpha) - R, & V &= F_s b - c, & & \\
K &= X(3c - 2X) + 2, & L &= bV. & &
\end{aligned}$$

In the plain chart let

$$E_c = k - R - 1 - q^4 s;$$

in the corrected chart let

$$E_c = 6(k - R - 1 - q^4 s) - v(R+1)(R-1)(R+3).$$

Finally define

$$\begin{aligned}
M &= hDH, \\
Z &= hD\{H(n - LB) + K\} - H(DB + hE_c), \\
P &= Q(Z^2 - M^2 G V^2) + (M^2 G - Z^2).
\end{aligned}$$

The requested single equation is $P = 0$. Every displayed abbreviation is a fixed polynomial. There are no variable exponents, divisions, factorials, or extra predicates in the emitted circuits.

**Theorem.** For either chart, ZFC proves

$$(\nexists J, F, \alpha, i, T, w, s, n, \eta, \zeta > 0 : \ P = 0) \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

### 8.3 Ordered soundness argument

For proof notation only, write

$$t = B/h, \quad W = 1 + hL, \quad \kappa = K/H, \quad \sigma = Y/q^3, \quad z = Z/M.$$

For the plain chart put $e = k - R - 1$, $\rho = e/q^7$. For the corrected chart put

$$\bar{e} = k - R - 1 - \frac{v(R+1)(R-1)(R+3)}{6}, \qquad \rho = \bar{e}/q^7.$$

No integrality of these quotients is initially assumed. Identically,

$$P = (Q-1)Z^2 - M^2 G(QV^2 - 1), \qquad z = n + \kappa + \sigma - Wt - \rho. \tag{F1}$$

**Root extraction and early positive transport.** At a positive zero, the identities

$$F_s = (1+2a)^2, \qquad Q = 4\Delta^2 U^2 G, \qquad \gcd(E, F_0) = 1$$

and prime valuations give $E = \tau^2$, $F_0 = f_0^2$, $G = g^2$, where $g = \tau f_0 > 0$. The retained rational adjacent-Pell descent gives a positive integer $y$ such that

$$QV^2 - (Q-1)y^2 = 1, \qquad z = \pm gy.$$

With $f = 1 + 2a$, $I = 2Ug$, one has

$$f^2 - \Delta I^2 = 1, \quad c^2 \mid I, \quad Q = (\Delta I)^2.$$

The signed folding theorem gives $c = \psi_A(p)$ for positive odd $p$. Here $\psi_A(j)$ denotes the Pell ordinate. Since $c \geq 3$, $p \geq 3$; therefore

$$c \geq 4A^2 - 1 > 4q^4.$$

The unconditional mask estimate is $R < q^4 + (\alpha+2)q^2$. Thus

$$b - c > \alpha(c - q^2) - q^4 - 2q^2 \geq c - q^4 - 3q^2 > 0.$$

This argument includes $Y = 1$ and does not assume a sign of $C$. The auxiliary Pell index is odd; because $V > c(f^2 - 1) > 4Q$, its first and third values are excluded. Hence

$$V > 8Q^2, \qquad b > 4g, \qquad gy < \frac{3}{8}bV < \frac{L}{2}. \tag{F2}$$

For the plain chart, $\rho < b/q^7 < L/2$. If $B \leq -1$, then $Wt < -L$, and (F1)–(F2) contradict $n > 0$. For the corrected chart the same proof applies whenever $(R+1)(R-1)(R+3) \geq 0$. Its remaining cases are $R \leq -4$ and $R = 0$. The exact identity

$$R = (hC + q\alpha)(q^2 - 1) + (M_C + qM_F)J$$

shows that, for $R = -r \leq -4$, $C < 0$, $|C| > r/q^3$, and $Wt < -(1 + w|C|)L$. The adverse correction is less than $ws^2r^3/(6q^4)$, whereas $L > (F_s - 1)r^2$ and $F_s - 1 > s^2$, so $w|C|L$ strictly exceeds it. For $R = 0$ the correction is only $ws^2/(2q^4)$ and the same margin suffices. Therefore both charts force $B \geq 0$, hence $C > 0$.

The calibrated mask bound is now licensed:

$$3q < R < q^4 - 2.$$

The normalized strong-rank and signed-index theorems, using $c^2 \mid I$, $c > 2R$, $c > 2p$, $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$, give $p = R$. In particular $R$ is odd.

**Authenticate the first index.** The first square gives $k = 2\psi_{1+2v}(n_0)$. Comparison of the two Pell bases gives $1 \leq n_0 < R$. Because $H \equiv -1 \pmod q$, clearing (F1) in the plain chart gives $q^4 \mid e$. The first recurrence gives $k \equiv 2n_0 \pmod{4v}$, and here $q^4 \mid v$. Consequently $q^4 \mid 2n_0 - R - 1$; its absolute value is less than $R < q^4$, so $2n_0 = R + 1$.

The corrected chart has only $q^3 \mid v$. Let $r_0 = (R+1)/2$. The exact Lucas expansion gives

$$2\psi_{1+2v}(m) = 2m + 8v\binom{m+1}{3} + 32v^2 Z_m, \qquad Z_m \in \mathbb{Z}_{\geq 0}.$$

Thus $\bar{e}$ is integral. Clearing (F1) and cancelling the literal factor 6 gives $q^7 \mid \bar{e} - q^4 s$, hence $q^4 \mid \bar{e}$. For $d = 2n_0 - R - 1$ one first obtains $q^3 \mid d$, and then

$$6(\bar{e} - d) = vd\{(2n_0)^2 + (2n_0)(R+1) + (R+1)^2 - 4\} + 192v^2 Z_{n_0}.$$

The right side is divisible by $q^6$. For every prime $\ell \mid q$, its consequence $v_\ell(\bar{e} - d) \geq 6v_\ell(q) - 1 \geq 4v_\ell(q)$ also covers $\ell = 2, 3$. Hence $q^4 \mid d$, and the same strict bound forces $d = 0$. Thus both charts give

$$n_0 = (R+1)/2. \tag{F3}$$

**Project before recovering the denominator.** Reduction of the literal numerator modulo $H$ gives $H \mid hK$ in the plain chart and $H \mid 6hK$ in the corrected chart. Only the power of $q$ was cancelled; no coprimality of $H$ and 3 is assumed.

Write $R = 2r+1$, $r \geq 48$. The retained ratio theorem is uniform for every $\lambda \geq 1$, with $A = \lambda Y(X+1)+1$ and first base $1 + 2X\lambda^2 Y^2$. Apply it at $\lambda = q$ and use (F3) and $Y < c/k < Y + 1$. It gives

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X+1) < H.$$

The universal projection therefore yields $H \mid d_0 h(X - 2^R)(2^R X + 1)$, with $d_0 = 1$ or 6. The entire product has absolute value below $H$:

$$q^3 h|(X - 2^R)(2^R X + 1)| < 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1}.$$

The last inequality holds for $X \geq 32, r \geq 48$; after division by $X^{r+1}$, the corner value is $2^{-117} + 2^{-25} < 1$, and both terms decrease. Consequently $X = 2^R$ and $\kappa$ is integral.

Now $q \mid X$ implies $q = 2^m$, $m \geq 5$, and $R > 3q > 6m$, so $q^6 \mid X$ and $q^8 \mid v$. For the plain chart (F3) gives $4v \mid e$; for the corrected chart it gives $\bar{e} = 32v^2 Z_{r_0}$. Thus $\rho$ is a nonnegative integer in both cases. Finally (F1) gives $Y/q^3 - WB/h \in \mathbb{Z}$. Reduction modulo $q^3$ and $h$ proves

$$q^3 \mid Y, \qquad h \mid B, \qquad B/h \geq 0.$$

The retained parity, exact half-binomial rounding and fresh native-source reconstruction now give a syntactic ZFC refutation.

### 8.4 Constructive converse and exact degree

A refutation supplies canonical outer data $q^3 \mid Y$, $X = 2^R$, $R \equiv 3 \pmod 4$, $R > 3q$, and $B/h \geq 0$. Set $s = Y$, and take $w = X/q^2$ or $w = X/q$ for the two charts. Use fresh bases at $\lambda = q$, and form $c = \psi_A(R)$, $k = 2\psi_{1+2v}((R+1)/2)$, $\eta = c - kY$, $\zeta = k - \eta$. The uniform ratio gives their strict positivity. A Pell matrix period modulo $c^2\tau$ supplies the seed $(f_0, I_0)$, and $i = I_0/(c^2\tau) > 0$. Double the seed as above. Positive auxiliary reconstruction gives $T_* > \alpha$, so $T = T_* - \alpha > 0$. Set

$$n = gy + (1 + hL)t + \rho - \kappa - Y/q^3.$$

This is positive because $g \geq f_0 > (A-1)I_0 > 2Xc$, whereas $\kappa < Xc$ and $Y/q^3 \leq Y < Xc$. Its carrier equals $gy$, proving $P = 0$.

For $q_0 = \beta J$, $k_0 = \eta + \zeta$, the complete leading forms are

$$[P_{\text{plain}}]_{312} = -2^{10} q_0^{140} w^{34} s^{78} i^{14} k_0^{44} (q_0 - F)^2,$$
$$[P_{\text{corrected}}]_{278} = -9 \cdot 2^{12} q_0^{106} w^{34} s^{78} i^{14} k_0^{44} (q_0 - F)^2.$$

Both have 135 terms. The highest part of $b$ comes from $-R$, not from $c(T+\alpha)$; retaining this change is essential. The independent checker proves 28 exact cut identities per circuit, covers every gate, verifies these entire leading forms, authenticates all coefficient bytes, and rejects twenty deliberate actual corruptions. Normal and optimized Python give identical mathematical receipts. The complete unbounded proof and retained ordinary dependencies are provided in the component's `PROOF.md` and source directory.

## 9 A cofinal Pell experiment that did not improve the endpoint

We also tested replacing the exact growth power by a cofinal Pell certificate. Put $d = 6(t+1)$ and impose

$$Y = d(3v+1), \qquad F = Y^2, \qquad X^2 - d(d+2)F = 1.$$

The nonzero ordinate of this Pell equation has an index at least $d$, because $Y_n(d+1) \equiv n \pmod{d}$. It follows that $F$ exceeds the factorial and label-growth budgets. A cofinal positive completion can use $n = d10^t$, provided the chosen horizon has $3 \nmid h$; then $9 \mid d$ and the Taylor congruence forces $Y_n(d+1)/d \equiv 1 \pmod 3$. Additional divisibility supplies a compatible beta base. The full argument and independent recurrence checks are in `experiments/cofinal_pell/`.

This alternative is mathematically valid but was *not* selected: in the matched compilation trial before the strict-horizon change, it cost two more variables than the shared exact-power route. This comparison is not against the later 236-variable endpoint. A first unqualified cofinal congruence claim also failed at $t = 16$; the preserved negative control detects that failure, and the corrected proof restricts the cofinal horizon. Documenting both facts avoids promoting a plausible compression idea before its arithmetic and actual endpoint cost are checked.

## 10 Use and incorporation

For a directly inspectable small-coefficient consistency equation, use (1). For the smallest number of integer variables among the new direct quartics, use the dense companion. For a compact straight-line expression in ten positive variables, use the corrected 89/278 chart, retaining its coefficient files and its distinct calibration. Replacing positive variables by $u+1$ changes the domain to nonnegative integers; it does not make an unrestricted-integer equation without additional work or a different construction.

From the archive root, the reproducibility entry points are:

```
python3 -B verify.py
python3 -B -O verify.py
python3 -B verify.py --rebuild
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The mathematical reconstruction uses only Python's standard library and no network. LaTeX is needed only to rebuild the report. The integration notes propose repository locations and explain the retained source boundary; no repository write or publication is part of this archive.

The proved progress is local and concrete: a full power certificate can be removed by changing the witness horizon convention, and a Pell scale can be moved to reduce the degree of a compact quotient construction. Stronger global claims about the simplest consistency equation would require a chosen complexity measure and further results.

# A Complete principal quadratic residuals

All variables $x_0, \dots, x_{235}$ range over unrestricted integers. The following 173 polynomials, together with (1), specify the complete equation. No additional predicates or side conditions are understood. Repeated subscripts mean ordinary multiplication or a displayed square. The readable source is generated from the literal JSON and checked against it by the release verifier.

$$R_{0} = x_{0} - x_{0}x_{2} - x_{2} + x_{7}$$

$$R_{1} = -x_{2}x_{5} - x_{3} + x_{6}x_{7}$$

$$R_{2} = -1 + x_{1} - x_{6}x_{7}$$

$$R_{3} = 1 + x_{3} - x_{4}x_{18} + x_{16}$$

$$R_{4} = -1 - x_{8}^{2} - 2x_{8}x_{9} - x_{9}^{2} + x_{19}$$

$$R_{5} = -10x_{8} + x_{8}^{2} - 6x_{8}x_{9} - 8x_{8}x_{19} - 20x_{9} - 11x_{9}^{2} - 16x_{9}x_{19} - 15x_{19} - 4x_{19}^{2} + x_{20}$$

$$R_{6} = -2 + 2x_{8} - x_{8}^{2} - 4x_{8}x_{20} - 4x_{20} - 4x_{20}^{2} + x_{21}$$

$$R_{7} = 1 + 6x_{8} - 36x_{8}^{2} - 24x_{8}x_{21} + 2x_{21} - 4x_{21}^{2} + x_{22}$$

$$R_{8} = -2 - x_{9} - x_{9}^{2} - 2x_{9}x_{10} - x_{10} - x_{10}^{2} + x_{23}$$

$$R_{9} = -8 + 7x_{8} - x_{8}^{2} + 4x_{8}x_{10} - 4x_{8}x_{23} - 4x_{10} + 4x_{23} + x_{24} - x_{46}$$

$$R_{10} = -1 + x_{8} - x_{8}^{2} - 2x_{8}x_{10} + x_{10} - x_{10}^{2} + x_{25}$$

$$\begin{aligned} R_{11} = {} & 23 + 16x_{8} - 8x_{8}x_{9} - 8x_{8}x_{10} - 16x_{8}x_{19} - 8x_{8}x_{25} + 19x_{9} + 9x_{9}^{2} - 36x_{9}x_{10} \\ & - 12x_{9}x_{25} + 16x_{10} + 8x_{10}^{2} - 24x_{10}x_{19} - 8x_{19}x_{25} - x_{20} + x_{26} - x_{31} \end{aligned}$$

$$R_{12} = -4x_{24}^{2} - 8x_{24}x_{26} - 4x_{26}^{2} + x_{27}$$

$$\begin{aligned} R_{13} = {} & -2 - 8x_{8} + 8x_{8}x_{23} + 8x_{8}x_{25} - 8x_{9} + 8x_{9}x_{23} + 8x_{9}x_{25} - 4x_{19} + 8x_{23} - 4x_{23}^{2} \\ & - 8x_{23}x_{25} + 8x_{25} - 4x_{25}^{2} + x_{28} \end{aligned}$$

$$\begin{aligned} R_{14} = {} & 6 + 4x_{8} + 32x_{8}x_{9} + 16x_{8}x_{19} + 16x_{8}x_{25} + 8x_{8}x_{28} + 24x_{9} - 32x_{9}x_{25} - 16x_{9}x_{28} \\ & + 12x_{19} - 16x_{19}x_{25} - 8x_{19}x_{28} - x_{20} + 8x_{25} - 16x_{25}^{2} - 16x_{25}x_{28} + 4x_{28} - 4x_{28}^{2} \\ & + x_{29} \end{aligned}$$

$$\begin{aligned} R_{15} = {} & 22 - 5x_{8} - x_{8}^{2} - 8x_{8}x_{9} - 12x_{8}x_{10} - 4x_{8}x_{23} - 8x_{9} - 16x_{9}x_{10} - 16x_{9}x_{23} - 20x_{10} \\ & - 20x_{10}^{2} - 24x_{10}x_{23} - 28x_{23} - 4x_{23}^{2} + x_{30} \end{aligned}$$

$$\begin{aligned} R_{16} = {} & 19 - 25x_{8} - 7x_{8}^{2} - 8x_{8}x_{9} - 30x_{8}x_{10} - 16x_{8}x_{25} + 3x_{9} + 5x_{9}^{2} - 4x_{9}x_{25} - 27x_{10} \\ & - 21x_{10}^{2} - 24x_{10}x_{25} - 6x_{23} - 17x_{25} - 4x_{25}^{2} + x_{31} \end{aligned}$$

$$R_{17} = 2 - 4x_{30}^{2} - 8x_{30}x_{31} - 4x_{31}^{2} + x_{32}$$

$$\begin{aligned} R_{18} = {} & -19x_{8} + 8x_{8}^{2} + 8x_{8}x_{19} + 8x_{8}x_{23} - 46x_{9} - 24x_{9}^{2} + 24x_{9}x_{19} + 24x_{9}x_{23} - 2x_{10} \\ & + 4x_{19} - 4x_{19}^{2} - 8x_{19}x_{23} + 14x_{23} - 4x_{23}^{2} - x_{25} + x_{33} \end{aligned}$$

$$R_{19} = -3 - 6x_{8} - 4x_{10} - 2x_{25} - 4x_{33} - 4x_{33}^{2} + x_{34}$$

$$\begin{aligned} R_{20} = {} & 8 + 42x_{8} + 7x_{8}^{2} + 28x_{8}x_{20} + 48x_{9} - 32x_{9}^{2} - 32x_{9}x_{19} + 64x_{9}x_{20} - 40x_{19}^{2} \\ & + 32x_{19}x_{20} + 14x_{20} - 4x_{20}^{2} + x_{35} + 2x_{38} \end{aligned}$$

$$R_{21} = 1 + 6x_{8} - 36x_{8}^{2} - 24x_{8}x_{35} + 2x_{35} - 4x_{35}^{2} + x_{36}$$

$$R_{22} = -x_{9}^{2} - 4x_{9}x_{36} - 4x_{36}^{2} + x_{37}$$

$$R_{23} = -6 - 2x_{8} - 4x_{9} - 14x_{19} - 16x_{19}^{2} + x_{38}$$

$$R_{24} = 24 + 21x_9 - 21x_9^2 - 18x_9x_{10} - 24x_9x_{23} - 24x_9x_{38} + 3x_{10} - 9x_{10}^2 - 16x_{10}x_{38} + 3x_{23} - 12x_{23}^2 - 8x_{23}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39} + 2x_{46}$$

$$R_{25} = 16 - 10x_8 - 20x_8^2 - 16x_8x_{10} - 24x_8x_{25} - 24x_8x_{39} - 12x_{10} - 16x_{10}x_{25} - 16x_{10}x_{39} - 14x_{25} - 4x_{25}^2 - 8x_{25}x_{39} + 4x_{39} - 4x_{39}^2 + x_{40}$$

$$R_{26} = -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{41}$$

$$R_{27} = 2 + x_8^2 - 4x_8x_{41} + 2x_{10} - 2x_{10}^2 - 8x_{10}x_{41} - 2x_{25} + 4x_{41} - 4x_{41}^2 + x_{42}$$

$$R_{28} = -31 - 16x_{10} - 16x_{10}x_{23} + 16x_{10}x_{25} - 8x_{23} + 12x_{23}^2 - 8x_{23}x_{25} + 24x_{25} - 4x_{25}^2 + x_{43} - 4x_{46}$$

$$R_{29} = -4x_{10} - 16x_{25}^2 - 32x_{25}x_{41} - 16x_{25}x_{42} - 16x_{25}x_{43} - 4x_{41} - 16x_{41}^2 - 16x_{41}x_{42} - 16x_{41}x_{43} - 2x_{42} - 4x_{42}^2 - 8x_{42}x_{43} - 4x_{43}^2 + x_{44}$$

$$R_{30} = -16x_8^2 + 8x_8x_{10} + 32x_8x_{25} + 16x_8x_{28} + x_{10} - x_{10}^2 - 8x_{10}x_{25} - 4x_{10}x_{28} - 16x_{25}^2 - 16x_{25}x_{28} - 4x_{28}^2 + x_{45}$$

$$R_{31} = -1 - 8x_{10} - 4x_{10}^2 + 8x_{10}x_{23} + 8x_{23} - 4x_{23}^2 + x_{46}$$

$$R_{32} = -1 - 9x_{10}^2 + 24x_{10}x_{23} + 12x_{10}x_{46} - 16x_{23}^2 - 16x_{23}x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = 1 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = -1 - 4x_{48} - 4x_{48}^2 + x_{49}$$

$$R_{35} = -1 - x_{10}^2 - 8x_{10}x_{48} - 4x_{10}x_{49} - 16x_{48}^2 - 16x_{48}x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{50} - 4x_{10}^2 - 8x_{10}x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = -4x_{45}^2 - 8x_{45}x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = -9 + 7x_8 - x_8^2 - 8x_8x_{45} - 4x_8x_{52} + 24x_{45} - 16x_{45}^2 - 16x_{45}x_{52} + 12x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = -1 - 4x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = -x_{10}^2 - 8x_{10}x_{53} - 4x_{10}x_{54} - 16x_{53}^2 - 16x_{53}x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{55} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{55} - 4x_{55} - 4x_{55}^2 + x_{56}$$

$$R_{42} = -10 - 12x_{56} - 4x_{56}^2 + x_{57}$$

$$R_{43} = 2 + 4x_{10} + 16x_{10}x_{56} + 8x_{10}x_{57} - 4x_{23} - 16x_{23}x_{56} - 8x_{23}x_{57} - x_{46} + 8x_{56} - 16x_{56}^2 - 16x_{56}x_{57} + 4x_{57} - 4x_{57}^2 + x_{58}$$

$$R_{44} = x_9 - x_9^2 + 8x_9x_{10} - 8x_9x_{23} - 4x_9x_{58} - 16x_{10}^2 + 32x_{10}x_{23} + 16x_{10}x_{58} - 16x_{23}^2 - 16x_{23}x_{58} - 4x_{58}^2 + x_{59}$$

$$R_{45} = -10 - 12x_{51} - 4x_{51}^2 + x_{60}$$

$$R_{46} = 1 - 16x_{51}^2 - 16x_{51}x_{59} - 16x_{51}x_{60} - 4x_{59}^2 - 8x_{59}x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = x_{10} - x_{10}^2 - 8x_{10}x_{59} - 4x_{10}x_{61} - 16x_{59}^2 - 16x_{59}x_{61} - 4x_{61}^2 + x_{62}$$

$$R_{48} = -6x_8 - 3x_8^2 - 18x_8x_9 - 16x_9 - 27x_9^2 + 4x_9x_{10} + 3x_{10} + 3x_{10}^2 - 4x_{10}x_{19} - 19x_{19} - 8x_{19}^2 + x_{20} + x_{63} - 4x_{76}$$

$$R_{49} = 1 - 8x_8 - 4x_8x_{10} + 8x_8x_{63} - 16x_9 - 8x_9x_{10} + 16x_9x_{63} + x_{10} + x_{10}^2 + 4x_{10}x_{19} + 8x_{19} - 8x_{19}x_{63} + 3x_{63} - 4x_{63}^2 + x_{64}$$

$$\begin{aligned}R_{50} = & -2-12x_8-16x_8x_{10}+16x_8x_{62}+16x_8x_{64}-24x_9-32x_9x_{10}+32x_9x_{62}+32x_9x_{64}\\ & +4x_{10}+4x_{10}^2+16x_{10}x_{19}+12x_{19}-16x_{19}x_{62}-16x_{19}x_{64}-4x_{62}^2-8x_{62}x_{64}-4x_{63}\\ & -2x_{64}-4x_{64}^2+x_{65}\end{aligned}$$

$$R_{51} = -2-4x_{65}-4x_{65}^2+x_{66}$$

$$R_{52} = -x_3x_{13}+x_{67}$$

$$R_{53} = x_{12}-x_{13}-2x_{22}x_{67}-x_{68}$$

$$R_{54} = x_{12}-x_{13}-4x_{24}x_{67}-2x_{27}x_{67}-x_{69}$$

$$R_{55} = -2x_{29}x_{67}+2x_{44}x_{67}-2x_{67}-x_{70}+x_{75}$$

$$R_{56} = x_{12}-x_{13}-4x_{30}x_{67}-2x_{32}x_{67}-x_{71}$$

$$R_{57} = x_{12}-x_{13}-2x_{34}x_{67}+6x_{67}-x_{72}$$

$$R_{58} = x_{12}-x_{13}-4x_{36}x_{67}-2x_{37}x_{67}-x_{73}$$

$$R_{59} = x_{12}-x_{13}-2x_{40}x_{67}-x_{74}$$

$$R_{60} = x_{12}-x_{13}-2x_{44}x_{67}-4x_{67}-x_{75}$$

$$R_{61} = -x_8x_9+x_8x_{10}+x_9+x_9x_{10}-x_{10}-x_{10}^2+x_{76}$$

$$R_{62} = 1+2x_{11}+x_{11}^2-x_{76}^2+x_{77}$$

$$R_{63} = -x_{77}^2+x_{78}$$

$$R_{64} = -4x_{65}x_{67}-2x_{66}x_{67}+2x_{67}+x_{79}$$

$$R_{65} = -x_{12}^2+2x_{12}x_{13}+2x_{12}x_{79}-x_{13}^2-2x_{13}x_{79}-x_{79}^2+x_{80}$$

$$R_{66} = -1-x_3x_9+x_{81}$$

$$R_{67} = -x_{13}x_{81}+x_{83}x_{84}+x_{85}$$

$$R_{68} = -1+x_3-x_3x_8+x_{82}$$

$$R_{69} = -1+2x_3+2x_3x_8+4x_3x_9-2x_3x_{19}+x_{83}$$

$$R_{70} = -x_{12}x_{82}+x_{84}$$

$$R_{71} = -6x_3x_{13}+x_{12}x_{81}+12x_{13}-8x_{13}x_{81}-4x_{13}x_{82}-x_{13}x_{83}+x_{86}$$

$$R_{72} = -x_{12}x_{68}+x_{13}x_{68}+x_{87}$$

$$R_{73} = -x_{69}x_{87}+x_{88}$$

$$R_{74} = -x_{70}x_{88}+x_{89}$$

$$R_{75} = -x_{71}x_{89}+x_{90}$$

$$R_{76} = -x_{72}x_{90}+x_{91}$$

$$R_{77} = -x_{73}x_{91}+x_{92}$$

$$R_{78} = -x_{74}x_{92}+x_{93}$$

$$R_{79} = -x_{75}x_{93}+x_{94}$$

$$R_{80} = -x_{85}x_{94}+x_{95}$$

$$R_{81} = -x_{86}x_{95}+x_{96}$$

$$R_{82} = -x_{96}^2+x_{97}$$

$$R_{83} = -x_2-x_2x_{16}+x_{16}+x_{98}$$

$$R_{84} = -x_{14}x_{98}+x_{99}$$

$$R_{85} = -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{99} + 2x_1x_{100} - x_{12}^2 - 2x_{12}x_{99} - x_{13}^2 - 2x_{13}x_{100} + x_{17}x_{102} - x_{78}x_{97} - x_{80}x_{97} - x_{99}^2 - x_{100}^2$$

$$R_{86} = -x_2x_{15} + x_{15} - x_{15}x_{98} + x_{100}$$

$$R_{87} = -x_8 - x_9 - 4x_{19} - 2x_{22} - 4x_{24} - 2x_{27} - 2x_{29} - 4x_{30} - 2x_{32} - 2x_{34} - 4x_{36} - 2x_{37} - 2x_{40} - 2x_{44} - 4x_{65} - 2x_{66} + x_{101}$$

$$R_{88} = -x_{105} - 4x_{105}x_{233} + x_{107}$$

$$R_{89} = -x_4x_{128} + x_{103}x_{225} - x_{106} - x_{107}x_{128}$$

$$R_{90} = x_{108}x_{225} - x_{109}x_{131} - x_{110}$$

$$R_{91} = x_{111}x_{225} - x_{112}x_{131} - x_{113}$$

$$R_{92} = x_{114}x_{225} - x_{115}x_{131} - x_{116}$$

$$R_{93} = x_{117}x_{225} - x_{118}x_{131} - x_{119}$$

$$R_{94} = x_{120}x_{226} - x_{122}x_{131} - x_{123}$$

$$R_{95} = x_{124}x_{226} - x_{125}x_{131} - x_{126}$$

$$R_{96} = -x_{17}x_{128} + x_{131}$$

$$R_{97} = x_{127} - x_{129}x_{131} - x_{130}$$

$$R_{98} = -x_{131}x_{133} + x_{132} - x_{134}$$

$$R_{99} = -x_{131}x_{136} + x_{135}x_{225} - x_{137}$$

$$R_{100} = -x_{131}x_{139} + x_{138}x_{225} - x_{140}$$

$$R_{101} = x_{141} - 2x_{141}x_{143} + x_{144}$$

$$R_{102} = -x_{141}x_{144} - x_{144} + x_{145}$$

$$R_{103} = 2x_{142} + x_{142}^2 + x_{144}^2 - x_{145}^2$$

$$R_{104} = -x_{142}x_{147} - x_{147} + x_{148}$$

$$R_{105} = 2x_{146} + x_{146}^2 + x_{147}^2 - x_{148}^2$$

$$R_{106} = 1 - 2x_0 + x_0^2 - 2x_0x_{142} + x_{149}$$

$$R_{107} = -1 - x_0x_{147} + x_3 + x_{142}x_{147} - x_{146} + x_{147} + x_{149}x_{150}$$

$$R_{108} = 1 - 2x_4 + x_4^2 - 2x_4x_{142} + x_{151}$$

$$R_{109} = -x_4x_{194} + x_{17} + x_{151}x_{152} - x_{193} + x_{195}$$

$$R_{110} = -1 + 2x_{142}x_{209} - x_{153} + 2x_{209} - x_{209}^2$$

$$R_{111} = 1 + x_{153}x_{154} - x_{193} - x_{194}x_{209} + x_{195} + 4x_{233}$$

$$R_{112} = -x_{142}x_{156} - x_{156} + x_{157}$$

$$R_{113} = 2x_{155} + x_{155}^2 + x_{156}^2 - x_{157}^2$$

$$R_{114} = 1 - 4x_{142} - 8x_{142}x_{233} + x_{158} + 8x_{233} + 16x_{233}^2$$

$$R_{115} = -1 + x_{103} - x_{155} - 2x_{156} - 4x_{156}x_{233} + x_{157} + x_{158}x_{159}$$

$$R_{116} = -1 + x_{104} - 2x_{142}x_{160} - x_{146} - x_{147} - 4x_{147}x_{233} + x_{148} + x_{158}x_{160} + x_{160} + 8x_{160}x_{233}$$

$$R_{117} = 1 + x_{142} - x_{142}x_{162} - x_{162} + x_{163}$$

$$R_{118} = 1 + 2x_{161} + x_{161}^2 - 2x_{162} + x_{162}^2 - x_{163}^2$$

$$R_{119} = 1 + x_{108} + x_{158}x_{164} - x_{161} - 2x_{162} - 4x_{162}x_{233} + x_{163} + 4x_{233}$$
$$R_{120} = -x_{142}x_{166} - x_{166} + x_{167}$$
$$R_{121} = 2x_{165} + x_{165}^2 + x_{166}^2 - x_{167}^2$$
$$R_{122} = -1 + x_{111} + x_{158}x_{168} - x_{165} - 2x_{166} - 4x_{166}x_{233} + x_{167}$$
$$R_{123} = -x_{142}x_{170} - x_{170} + x_{171}$$
$$R_{124} = 2x_{169} + x_{169}^2 + x_{170}^2 - x_{171}^2$$
$$R_{125} = -1 + x_{114} + x_{158}x_{172} - x_{169} - 2x_{170} - 4x_{170}x_{233} + x_{171}$$
$$R_{126} = -x_{142}x_{174} - x_{174} + x_{175}$$
$$R_{127} = 2x_{173} + x_{173}^2 + x_{174}^2 - x_{175}^2$$
$$R_{128} = -1 + x_{117} + x_{158}x_{176} - x_{173} - 2x_{174} - 4x_{174}x_{233} + x_{175}$$
$$R_{129} = -x_{142}x_{178} - x_{178} + x_{179}$$
$$R_{130} = 2x_{177} + x_{177}^2 + x_{178}^2 - x_{179}^2$$
$$R_{131} = -1 + x_{120} + x_{158}x_{180} - x_{177} - 2x_{178} - 4x_{178}x_{233} + x_{179}$$
$$R_{132} = -x_{142}x_{182} - x_{182} + x_{183}$$
$$R_{133} = 2x_{181} + x_{181}^2 + x_{182}^2 - x_{183}^2$$
$$R_{134} = -1 + x_{121} - 2x_{142}x_{184} + x_{158}x_{184} - x_{181} - x_{182} - 4x_{182}x_{233} + x_{183} + x_{184} + 8x_{184}x_{233}$$
$$R_{135} = -x_{142}x_{186} - x_{186} + x_{187}$$
$$R_{136} = 2x_{185} + x_{185}^2 + x_{186}^2 - x_{187}^2$$
$$R_{137} = -1 + x_{124} + x_{158}x_{188} - x_{185} - 2x_{186} - 4x_{186}x_{233} + x_{187}$$
$$R_{138} = -x_{142}x_{190} - x_{190} + x_{191}$$
$$R_{139} = 2x_{189} + x_{189}^2 + x_{190}^2 - x_{191}^2$$
$$R_{140} = -1 + x_{127} + x_{158}x_{192} - x_{189} - 2x_{190} - 4x_{190}x_{233} + x_{191}$$
$$R_{141} = 1 - x_{142}x_{194} - x_{194} + x_{195}$$
$$R_{142} = -1 + 2x_{193} + x_{193}^2 + x_{194}^2 - 2x_{195} - x_{195}^2$$
$$R_{143} = x_{128} - 2x_{142}x_{196} + x_{158}x_{196} - x_{193} - x_{194} - 4x_{194}x_{233} + x_{195} + x_{196} + 8x_{196}x_{233}$$
$$R_{144} = -x_{142}x_{198} - x_{198} + x_{199}$$
$$R_{145} = 2x_{197} + x_{197}^2 + x_{198}^2 - x_{199}^2$$
$$R_{146} = -1 + x_{132} + x_{158}x_{200} - x_{197} - 2x_{198} - 4x_{198}x_{233} + x_{199}$$
$$R_{147} = -x_{142}x_{202} - x_{202} + x_{203}$$
$$R_{148} = 2x_{201} + x_{201}^2 + x_{202}^2 - x_{203}^2$$
$$R_{149} = -1 + x_{135} + x_{158}x_{204} - x_{201} - 2x_{202} - 4x_{202}x_{233} + x_{203}$$
$$R_{150} = -x_{142}x_{206} - x_{206} + x_{207}$$
$$R_{151} = 2x_{205} + x_{205}^2 + x_{206}^2 - x_{207}^2$$
$$R_{152} = -1 + x_{138} + x_{158}x_{208} - x_{205} - 2x_{206} - 4x_{206}x_{233} + x_{207}$$

$$\begin{aligned}R_{153} = &-7-2x_0^2-2x_0x_3-x_1^2-x_2^2-2x_3^2-x_4^2+2x_8+2x_8x_9-x_{10}^2-x_{11}^2\\&-x_{12}^2-x_{13}^2-x_{14}^2-x_{15}^2-x_{16}^2-x_{17}^2-x_{19}-x_{101}^2-x_{103}^2-x_{104}^2\\&-x_{108}^2-x_{111}^2-x_{114}^2-x_{117}^2-x_{120}^2-x_{121}^2-x_{124}^2-x_{127}^2-x_{128}^2\\&-x_{132}^2-x_{135}^2-x_{138}^2+x_{141}-x_{209}^2-24x_{233}-32x_{233}^2\end{aligned}$$

$$\begin{aligned}R_{154} = &-x_4-x_4x_8-x_4x_9-x_4x_{10}-x_4x_{11}-x_4x_{12}-x_4x_{13}-x_4x_{14}-x_4x_{15}\\&-x_4x_{16}+x_{209}\end{aligned}$$

$$R_{155} = -16+x_0-x_{210}^2$$

$$R_{156} = -x_{142}x_{212}-x_{212}+x_{221}$$

$$R_{157} = -1+x_{211}^2+x_{212}^2-x_{221}^2$$

$$R_{158} = -x_{214}-x_{214}x_{215}+x_{222}$$

$$R_{159} = -1+x_{213}^2+x_{214}^2-x_{222}^2$$

$$R_{160} = -x_{206}^2+x_{223}$$

$$R_{161} = x_{212}-2x_{217}x_{223}-x_{223}$$

$$R_{162} = -4x_{212}-8x_{212}x_{216}+x_{215}$$

$$R_{163} = -x_{142}-x_{211}x_{218}+x_{215}$$

$$R_{164} = -1-x_{205}-x_{211}x_{219}+x_{213}$$

$$R_{165} = -x_{101}-4x_{206}x_{220}+x_{214}$$

$$\begin{aligned}R_{166} = &-2+2x_0^2+2x_0x_3-2x_0x_{147}-2x_0x_{156}+x_1^2-2x_1x_{194}+x_2^2-2x_2x_{182}+x_3^2\\&-2x_3x_{156}+x_8^2-2x_8x_{162}+x_9^2-2x_9x_{166}+x_{10}^2-2x_{10}x_{170}+x_{11}^2-2x_{11}x_{174}\\&+x_{12}^2-2x_{12}x_{178}+x_{13}^2-2x_{13}x_{186}+x_{14}^2-2x_{14}x_{190}+x_{15}^2-2x_{15}x_{198}+x_{16}^2\\&-2x_{16}x_{202}-2x_{101}x_{206}-x_{142}x_{224}+x_{146}^2+x_{155}^2+x_{161}^2+2x_{162}+x_{165}^2\\&+x_{169}^2+x_{173}^2+x_{177}^2+x_{181}^2+x_{185}^2+x_{189}^2+x_{193}^2+x_{197}^2+x_{201}^2\\&+x_{205}^2+2x_{206}^2-x_{227}^2-x_{228}^2-x_{229}-x_{229}^2\end{aligned}$$

$$R_{167} = -x_{104}x_{225}+x_{128}$$

$$R_{168} = -x_{121}x_{226}+x_{128}$$

$$R_{169} = -1+x_2-x_3x_{234}$$

$$\begin{aligned}R_{170} = &-2-x_{101}^2-x_{147}^2-x_{156}^2+2x_{162}-x_{162}^2-x_{166}^2-x_{170}^2-x_{174}^2-x_{178}^2\\&-x_{182}^2-x_{186}^2-x_{190}^2-x_{194}^2-x_{198}^2-x_{202}^2+x_{206}^2-x_{227}^2-x_{228}^2\\&-x_{229}-x_{229}^2\end{aligned}$$

$$\begin{aligned}R_{171} = &-1-4x_{106}^2-4x_{110}^2-4x_{113}^2-4x_{116}^2-4x_{119}^2-4x_{123}^2-4x_{126}^2+x_{128}^2-4x_{130}^2\\&-4x_{134}^2-4x_{137}^2-4x_{140}^2-x_{230}^2-x_{231}^2-x_{232}-x_{232}^2\end{aligned}$$

$$R_{172} = -1+x_{234}-4x_{235}-4x_{235}^2$$

## B Artifact identities and comparison scope

| Object | Relative path |
|---|---|
| Principal residuals | `direct/principal236/equation.json` |
| Principal full quartic | `direct/principal236/expanded_polynomial.json` |
| Dense residuals | `direct/dense235/equation.json` |
| Dense full quartic | `direct/dense235/expanded_polynomial.json` |
| Strict arithmetic source | `direct/source/raw_source.json` |
| Plain compact circuit | `compact/equation_83_degree312.json` |
| Corrected compact circuit | `compact/equation_89_degree278.json` |
| Exact fixed compact tuple | `compact/upstream_lucas/parent/coefficients/` |
| Proof dependencies | `sources/factorial_free/` and `compact/` |
| Independent reviews and receipts | `review/` |

The principal residual JSON has SHA-256

`4e619e431c9beb13967aff3e7e7db35c56951fb179c353c43ee1fe7601dc3cbe.`

The dense residual JSON has SHA-256

`9988b2f76ef9f2adb4d7f6f9867134b1dc695905c0fc1aa189da753a745949d4.`

The root source ledger records inspected ZIP identities and the repository snapshot; `MANIFEST.sha256` fixes the delivered files. The comparison covers the linked repository directories and their incoming submissions at that snapshot. It does not establish a universal or historical optimum.

## References

[1] Vladimir Reshetnikov and contributors, *ProveIt*, repository snapshot `7a8da414376a80b90ecf799f37ceb26fb9ac674e`, https://github.com/VladimirReshetnikov/ProveIt/tree/7a8da414376a80b90ecf799f37ceb26fb9ac674e. The inspected directories are `Computability/HilbertTenthProblem`, `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, and `docs/incoming`.

[2] Incoming ProveIt submission, *Removing Factorial Authenticity: Binomial Prime-Power Certificates for Diophantine Consistency*, 25 October 2026, `ConZFC_Factorial_Free_Binomial_CRT_2026-10-25.zip`. Complete supplied source retained under `sources/factorial_free/`.

[3] Incoming ProveIt submission, *Cofinal Witnesses and Centered Carriers*, 25 October 2026, `ConZFC_Cofinal_Witnesses_Centered_Carriers_2026-10-25.zip`. Relevant compact and direct dependencies are retained in their source chains.

[4] Incoming ProveIt submission, *Normalized Residues, Round 19*, 25 October 2026, `ConZFC_Normalized_Residues_Round19_2026-10-25.zip`. Retained within the direct dependency tree.

[5] Incoming ProveIt submission, *Shared Factorial Error*, 25 October 2026, `ConZFC_Shared_Factorial_Error_2026-10-25.zip`. Source of the signed-binomial-digit comparison used here.

[6] Incoming ProveIt submission, *Lucas Corrections and Witness Elimination*, 25 October 2026, `ConZFC_Lucas_Corrections_and_Witness_Elimination_2026-10-25.zip`. Retained under `compact/upstream_lucas/`.

[7] ProveIt logical calibration, pinned source commit `08200eb10fde2db408c69dffd840d6920a569b1f`. See `KEndpoint.lean`, `ConstructibleChoice.lean`, and `CALIBRATION_PINS.json` in `sources/factorial_free/direct/parent/upstream/round17/architecture/proofs/inherited/calibration/`. The pins identify theorem statements and source provenance; the present release does not claim a new full build of their dependency graph.